

Car Parking Sensor

(Parking Assistant)

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1. Problem

The goal of this project was to build a small but non-trivial prototype (sensor + actuator) in the Tinkercad virtual Arduino environment. This project is a **car parking sensor** which measures the distance to an obstacle and warns the driver when driver reverses towards an obstacles with LEDs, a buzzer and an LCD display. The closer the obstacle, the stronger the beeps and warning.

2. Design

An ultrasonic sensor sends a pulse and measures the time until the echo return. The Arduino converts this time to centimeters:

$$d = \frac{t \cdot 0.0343}{2}$$

where t is the echo time in microseconds.

2.1. Distance zones

The measured distance selects one of four zones, which determines LED, sound and text on the LCD (Table 1).

Table 1: Distance zones and device reaction

Distance	LED	Buzzer	LCD
> 100 cm	Green	Silent	Distance in cm
50-100 cm	Yellow	Slow beeping (1000 Hz beep, 500 ms pause)	Distance in cm
25-50 cm	Red	Fast beeping (1500 Hz beep, 500 ms pause)	Distance in cm
< 25 cm	Red	Almost continuous (2000 Hz beep, 200 ms pause)	"STOP!!"

2.2. Program structure

The program is split into small functions:

- `readDistanceCM()` - triggers the sensor and returns the distance.
- `setLEDS()` - turns the green, yellow and red LEDs on or off.
- `showDistanceLCD()` - function that writes to the LCD screen.
- `setup()` and `loop()` - initialisation and the main cycle.

Note. The main cycle repeats continuously.

3. Parts lists

Table 2 shows all components used in this project.

Table 2: Parts list

Name	Quantity	Component
U1	1	Arduino Uno R3
1	1	Breadboard Small
PING1	1	Ultrasonic Distance Sensor
D1	1	Green LED
D2	1	Yellow LED
D3	1	Red LED
PIEZO1	1	Piezo
R1 R2 R3	3	220 Ω Resistor
R4	1	110 Ω Resistor
U2	1	PCF8574-based, 32 (0x20) LCD 16x2 (I2C)

4. Circuits

4.1. Wiring circuit and pin connections

Figure 1 shows the circuit built in Tinkercad and Table 3 lists the pin connections.

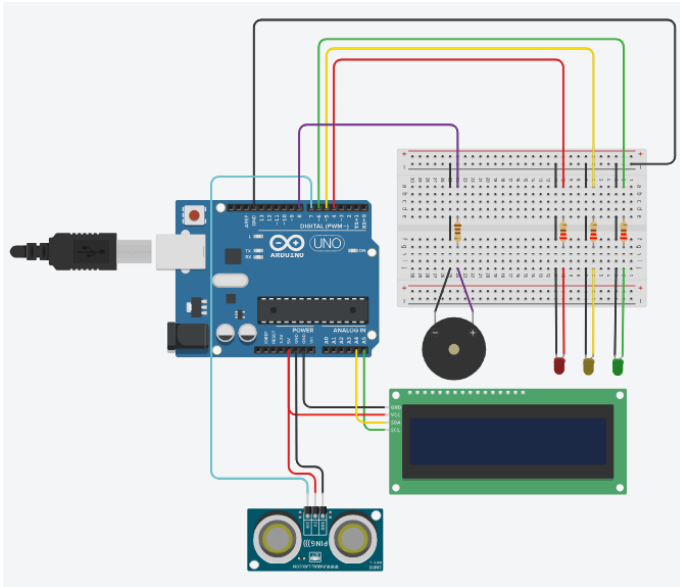


Figure 1: Wiring in Tinkercad

Table 3: Pin connections

Component	Arduino pin
Green LED (through 220 Ω)	D6
Yellow LED (through 220 Ω)	D5
Red LED (through 220 Ω)	D4
Ultrasonic sensor SIG	D7
Piezo buzzer (+) (through 110 Ω)	D8
LCD SDA / SCL	A4 / A5
Sensor, LCD VCC	5V
LED cathodes, buzzer (-), sensor and LCD GND	GND

4.2. Schematic

Figure 2 shows the circuit generated in Tinkercad.

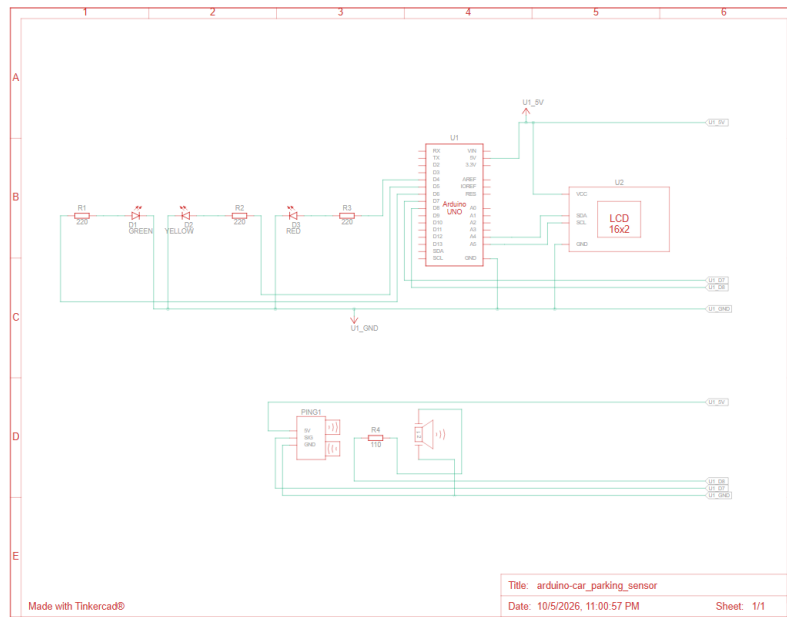


Figure 2: Schematic generated by Tinkercad

5. Results

5.1. What worked

This list below shows, what worked during this project:

- Distance measurement with the ultrasonic sensor.
- LEDs, buzzer and LCD react correctly to all four zones.
- Splitting the code into functions for better/easier readability.

5.2. What did not work (and how it was fixed)

Table 4 lists the problems, cause and how they were solved.

Table 4: Problems encountered during development

Problem	Cause	Solution
LEDs did not light	LED polarity was reversed	Flipped the LEDs (anode to resistor, cathode to GND)
Wrong LED colours	Pin numbers in the code did not match the wiring	Corrected the pin numbers in the code
LCD flickered between "STOP!!" and "distance"	Two places wrote to the same LCD row at the same time	Write to the LCD from one function only

6. Future improvements

To improve this project, next features should be made:

- Make beeping smoother.
- More sensors (left, center, right) to show where the obstacle is.
- Building it with real hardware.
- Use LCD distance bar instead of decimal distance.
- Add a push button which turns the buzzer off.
- Use PWM to fade the LED with distance instead of three fixed zones.

7. Conclusion

The prototype works as a simple parking assistant. Components such as ultrasonic sensor measures the distance and the LEDs, buzzer and LCD screen react to it in real time.

8. Source code

The full code is available in the repository : https://github.com/dzenardas/arduino-car_parking_sensor

9. Demo video

Watch the demo on YouTube: <https://youtu.be/-p2tUF-fARI>